18

PHONOLOGICAL ACQUISITION OF PORTUGUESE AS A HERITAGE LANGUAGE

Anabela Rato

This chapter surveys the state of research on the phonological acquisition of Portuguese as a heritage language, discusses the main topics in the field of heritage speech acquisition, and provides directions for future research. The chapter begins with a definition of heritage language (HL) and heritage speaker (HS) and a description of the different profiles of heritage speakers of Portuguese that differ in relation to their experience with the heritage language. It overviews the most relevant findings in heritage Portuguese speech research with respect to the domains of speech perception and production, and it presents an account of the still scarce research on the phonological acquisition of Portuguese as a heritage language. Finally, it identifies the major empirical gaps and outlines directions for future investigation.

1 Introduction

This chapter presents an overview of the existing research on the phonological acquisition of Portuguese as a heritage language (HL) and identifies the production and perceptual patterns in heritage Portuguese speech. It, therefore, surveys the state of the science on heritage Portuguese speech acquisition and discusses the main research questions in the field.

In heritage language acquisition, the domains of phonetics and phonology are somewhat less investigated in comparison with other linguistic areas such as morphology and syntax. One of the possible reasons for this is because heritage speakers (HS) generally sound native-like, even if they have a slight 'accent' (e.g., Polinsky & Kagan 2007), as perceived by monolingual native speakers. In the case of Portuguese, the same trend is observed. Research is very scarce, and little is known about the nature of phonological representations in heritage Portuguese speakers. Thus, this chapter also aims to identify the research gaps in heritage Portuguese phonological acquisition as well as provide directions for future investigation.

This section begins with a definition of heritage language and heritage speaker and a description of the profiles of HSs of Portuguese that differ in relation to their experience with the HL.

Rothman (2009: 156) defines heritage language as "a language spoken at home or otherwise readily available to young children, and crucially this language is not a dominant language of

DOI: 10.4324/9781003266648-22

the larger (national) society." Heritage speakers are, thus, early bilinguals whose home language (i.e., a minority language in many cases) coexists with the societal language (i.e., the majority language) in a multilingual context. The language spoken at home is generally the first language (L1) to which HSs are exposed and it is acquired through interaction with naturalistic input and the linguistic mechanisms involved in child language acquisition (Rothman 2009). Although a HL is acquired early on, the exposure to this language is generally limited to certain social contexts, such as the home environment and the local migrant community. Contact with the societal language may start at birth if also spoken within the family or it gradually increases as the bilingual child starts to socialize outside the home and goes to kindergarten or school (Montrul 2010; Rothman 2009). The exposure to the HL tends to progressively reduce when the bilingual child starts to have more contact with the majority language, in contexts where it is the main or only language of instruction. In any of these circumstances, the heritage language is the L1, one of two L1s (2L1), or two of three L1s (3L1) in cases where each parent or caregiver speaks a different home language. Heritage speakers are, thus, either simultaneous or successive bilinguals with the majority language as an early L2. Chang (2021: 581) further defines HSs as "language users that share the core characteristics of an 'interrupted' trajectory of exposure to their first language (L1), where the discontinuity is brought about by intensive exposure to a second language (L2)." Although there are diverse populations that fall within this definition (e.g., international adoptees, forced migrants, or volitional migrants), the heritage speaker profile of the research under discussion is the most common "switched-dominance bilinguals" (Chang 2021: 582), i.e., those who had continued contact with their HL during childhood within migratory contexts but experience a language shift, becoming L2-dominant bilingual speakers (Flores et al. 2022). As pointed out by Kim (forthcoming), these specific conditions of language acquisition, including a switch of language dominance, an asymmetry between language input and use, and acquisition of 'diasporic' varieties, promote a particular development of the heritage language that may differ in some linguistic domains from the homeland language and a unique acquisition context that helps further our understanding about heritage speech acquisition. However, it is important to note that underlying HL development there is a high degree of heterogeneity in the acquisition process; for instance, in level of exposure and language use, and contexts that result in different profiles of HSs. These range from highly proficient HL speakers ('talkers') who sound like their homeland counterparts to receptive bilinguals ('listeners', 'overhearers') who may understand their HL but do not speak it (Montrul 2016). It is relevant to note that in the field of sociolinguistics, heritage speakers are defined by their 'linguistic autobiography' (Nagy 2015: 310), that is, the focus is on their ethnic identity, cultural ties, and other social factors and less on their HL proficiency and fluency.

In the case of Portuguese, a heritage speaker is typically a second- or third-generation migrant whose parents are from a Portuguese-speaking[1] country and who learned the language from birth or from early on within the family. Portuguese HSs may also have contact with the minority language in the larger local migrant Portuguese-speaking community, in heritage language classes, and in short-stay visits to the homeland. Several studies on the acquisition of different linguistic properties have been carried out about Portuguese HSs living in non-Portuguese-speaking countries, specifically on European Portuguese (EP) or Brazilian Portuguese (BP) speakers living in Europe (e.g., Andorra: Rinke & Flores 2018; Finland: Piippo 2016; Ribeiro 2021; France: Flores et al. 2022, 2016; Tomaz et al. 2019, 2020; Germany: Jochmann 2020; Rinke & Flores 2021; Luxembourg: Engel de Abreu et al. 2012; Switzerland: Flores et al. 2022; the United Kingdom: Ferreira 2020), in North and South America (Canada: Rato & Machado 2024; Scetti 2016; Simard et al. 2013; United States: Osborne 2021; Pires & Rothman 2009; Silva & Soares 2021; Venezuela: Nunes 2018), or in Africa (South-Africa: Pereira 2017). A survey of this recent body of work con-

firms the general tendency observed in heritage language acquisition, with much sparser research in the acquisition of phonetics and phonology than in other linguistic domains such as morpho-syntax.

To my knowledge there are also no linguistic studies to date that focus on migrant families originally from Portuguese-speaking African or Asian countries who speak Portuguese as their home language, although there is significant migration from these countries to non-Portuguese-speaking societies, e.g., Cape Verdean communities in the United States (Halter 2005) and Angolan communities in Canada (Lebert 1999). This may be explained by the particular position of the Portuguese language in the Portuguese-speaking African countries of Angola, Cape Verde, Equatorial Guinea, Guinea-Bissau, Mozambique, and São Tomé and Príncipe, as well as in East Timor and Macau (China), where the language is, for many families, the official societal language but not the language learned or spoken at home. For example, Mozambican speakers are primarily exposed to one of the many Mozambican Bantu languages as a family language and, within this scenario, in situations of migration, Portuguese may coexist (or not) with the family language, which will be transmitted as HL.

An inverse situation is that of the acquisition of native African languages as HL in Portuguese-speaking societies. This is the case, for instance, of Cape Verdean Creole in Portugal (e.g., Costa et al. 2016). In this context, Portuguese is the dominant majority language and not a HL. A similar situation is that of Germanic language enclaves in Brazil, where German dialects coexist over several generations as HLs with Portuguese, the majority language, since the 19th century (e.g., Altenhofen 1996; Flores et al. 2022; Koch 1974).

Another profile of HSs of Portuguese is that of the 'returnees' (Flores, 2010; Flores & Rato 2016; Flores & Snape 2021; Koven 2007). Returnees are second- or third-generation speakers who move to their parents' homeland (in this case, a Portuguese-speaking country) after spending childhood/adolescence in the host country. Although moving to the homeland is not a 'return'—since most often they were born in the host country—it can be considered a situation of remigration because it is a change to their nuclear family's country of origin (i.e., the heritage country). To date, research on returnee Portuguese speakers is mainly focused on those from German-speaking countries (e.g., Flores & Rato 2016) and on the language that ceases to be the environmental language (e.g., Flores 2010). Only a few studies have analyzed the development of the heritage language after the return to the homeland environment (e.g., Flores & Rato 2016; Flores et al. 2022).

Bilingual speakers who, though sharing the same language, acquire two different varieties of Portuguese, termed 'bilectal bilinguals' (cf. Castro et al. 2017; Figueira 2017; Kupisch et al. 2023) have been the target population of studies on the European Portuguese (EP)–Brazilian Portuguese (BP) dyad. These are, for example, Brazilians who immigrate to Portugal or Portuguese who move to Brazil and are exposed to the two language varieties either as children or as adults (i.e., early vs. late bilectal bilinguals). The specific contact situation of two typologically similar national varieties with a high degree of mutual intelligibility but which differ in many microparametric domains (e.g., Martins 2006), including their sound system (e.g., Escudero et al. 2009), raises questions regarding the organization of the two dialects in a bilectal bilingual's mind, the ease/difficulty of keeping the two linguistic systems separate, and the degree of cross-linguistic influence bidirectionally. The study of bilectal bilingualism involving closely related varieties provides a particular contact situation to investigate whether a high degree of typological proximity facilitates or hinders heritage language acquisition and maintenance.

Though far from being exhaustive, the examples provided so far aimed to describe the main profiles of heritage speakers and of language contact situations in which Portuguese is acquired

as a heritage language. The study of the acquisition of heritage Portuguese speech is therefore of interest due to the linguistic specificities underlying the acquisition of the HL in a migration context. These conditions result from the fact that, although HSs tend to be exposed to their HL from birth, early exposure is constrained by reduced contact with their HL language and by the massive exposure to a societal dominant language.

We turn now to the main research questions in the field of phonological acquisition of heritage languages, with a focus on the findings of theoretically driven empirical research on Portuguese.

2 Phonological acquisition of Portuguese as a heritage language

Although the phonological and phonetic competence of heritage speakers tends to be a well-preserved aspect of linguistic proficiency in heritage language acquisition, with heritage speakers sounding very much like native speakers, research has also shown that they can be identified by monolingual homeland speakers as not exactly 'native'. Research on HSs speech has also observed that, though their accent sounds less native-like than monolingual speakers with more variation in accent ratings, it sounds more native-like than second language learners (L2ers), which places heritage speakers with a phonetic advantage.

The findings of a large body of work in different dominant-heritage language pairings have also revealed that the production and perceptual performance of HSs differ from that of monolingual native speakers with certain phonetic patterns in their speech diverging from those of the homeland speakers (Flores & Rato 2016; Flores et al. 2017; Hrycyna et al. 2011; Kim 2012, 2015; Nagy & Kochetov 2013; Rao & Ronquest 2015; Ronquest 2013). Specifically, heritage speakers show variation regarding perceived accent (Kupisch et al. 2014; Rato et al. 2015; Stangen et al. 2015), speech production (Chang et al. 2011; Godson 2004; Rao 2015) and speech perception (Chang 2016; Gor 2014; Lukyanchenko & Gor 2011), and both at the segmental level (Au & Oh 2009; Baker & Trofimovich, 2005; Godson 2004; Guion 2003) and at the prosodic level (Chang & Yao 2016; Colantoni et al. 2016; Kim 2020; Yang 2015). Although the emergent research on the phonetic and phonological competence and performance of heritage speakers has shown that, on the one hand, the early exposure to the family language is a strong predictor of a native-like accent, several studies have also found differences between heritage and homeland speakers regarding their speech. Since the minority language is more subject to input changes and variation in language use, it is in the heritage language (and less in the majority language) that HSs tend to develop a distinct sound system (Kupisch et al. 2020). The acquisition of the heritage phonological system has been investigated mostly with global measures such as accent, comprehensibility and intelligibility ratings, acoustic data of speech production, and/or speech perception categorizations, such as identification and discrimination of phonemes and prosodic patterns.

The rest of this chapter is, thus, organized as follows. In §2.1, I review the research on holistic measures of heritage speech, and in §2.2 I survey the research on speech production and perception of segments and suprasegments. Both sections comprise a discussion of the main research questions in the field of heritage speech acquisition that have been addressed in several language pairings with a focus on the existing research on the acquisition of heritage Portuguese speech.

2.1 Heritage accent

As previously mentioned, it is often acknowledged that heritage speakers sound 'native', i.e., they sound similar to homeland speakers when using the heritage language. This is not surprising given that the sound system of the heritage language is acquired early on due to early language exposure. A heritage speaker with early and continuous contact with the HL during childhood is predicted to

have a native homeland-like linguistic competence in domains that are acquired early in native language development, including at the phonological and phonetic levels (Benmamoun et al. 2013; Montrul 2016). Studies comparing different linguistic abilities have found that the same heritage speakers show more native-like performance in phonology and phonetics than in morphosyntax, in which more deviations from the baseline are reported (Montrul 2016; Polinsky 2018). Research findings on the global perception of heritage speech have systematically corroborated impressionistic observations that heritage speakers tend to develop an accent that is similar to that of native peers living in the homeland and an apparent advantage in phonological categorization and perception (Polinsky 2018), which may have led to the scarceness in research in this domain and generally fewer studies on the phonological acquisition of heritage speakers in comparison with other linguistic domains (Montrul 2016; Polinsky & Kagan 2007). Kim (forthcoming) also points out other aspects that contribute to the acknowledgment of this advantage in speech performance. In her review, Kim includes studies that show a lasting effect of early heritage language contact on HSs speech production not only when using the HL regularly (Au et al. 2008; Oh et al. 2003), but also in cases in which HSs did not speak the language regularly during childhood (Au et al. 2002, 2008; Knightly et al. 2003). However, impressionistically, it is also noted that HSs sound different from homeland speakers, leading to what has been termed a 'heritage' accent (Benmamoun et al. 2010, 2013; Chang 2021; Polinsky 2018; Rao 2015). Research has provided evidence that heritage speakers sound less native than homeland speakers and more native than late L2 learners (Au et al. 2002, 2008; Flores & Rato 2016; Knightly et al. 2003; Kupisch et al. 2014, 2020; Lloyd-Smith et al. 2020; Oh et al. 2003). Heritage speech is also assessed with varying accent ratings and with higher degrees of uncertainty than homeland speakers' and late L2 learners' (Chang & Yao 2016; Flores & Rato 2016; Kan 2020; Kupisch et al. 2014, 2020; Lloyd-Smith et al. 2020; Stangen et al. 2015). Research on the perception of global accent in heritage speech has thus found that exposure to a language since birth may not be sufficient for the development and/or maintenance of homeland-like speech, even when speakers maintain contact with the L1 during adulthood (De Leeuw et al. 2010; Godson, 2004; Kupisch et al. 2014; Lein et al. 2015; Stangen et al. 2015; Vago 1991). The massive exposure to the societal language from early on may lead to contact-induced changes in the HL sound system and, consequently, to the development of accented speech, despite early exposure (Godson 2004; Kupisch et al. 2014; Stangen et al. 2015). Although research has been mostly focused on adult bilingual populations, a few studies examining child heritage accent (Correia et al. 2022; Kupisch et al. 2021; Wrembel et al. 2019) also show that children speak their heritage language with a noticeable accent. The findings reported by Kupisch and colleagues (2021) suggest that changes in the degree of accentedness are associated with language dominance shift. However, the developmental trajectory of a heritage accent, that is, when it starts to surface and how it changes over time, is not yet fully understood. Studies on L1 attrition have shown that these changes may even occur when immersion in a new linguistic environment takes place in adulthood and lasts over decades (De Leeuw et al. 2010; Hopp & Schmid 2013), which may indicate that the effects of the dominant language may extend over a lifetime. Therefore, the sources of heritage accent, the interaction between the sound systems of the heritage language and the dominant language, and the role of the factors impacting the speech performance of heritage languages are topics that need further examination. Although early exposure seems to contribute to the development of a robust phonological system, the speech of heritage speakers exhibits considerable variation. Research findings on different heritage-dominant dyads suggest that the influence of the dominant language and the gradually decreasing exposure to and use of the minority language may shape the heritage speech with non-native features (e.g., Au et al. 2002; Godson 2004; Hopp & Schmid 2013; Oh et al. 2003), contributing to the perception of a heritage accent (Benmamoun

et al. 2010, 2013). Still, despite the individual variability perceived in their speech, heritage speakers consistently outperform L2 learners, demonstrating the advantage of the early exposure to the language (e.g., Au et al. 2002; Flores & Rato 2016; Oh et al. 2003, Rato et al. 2015). The role of different individual factors, such as heritage language experience during childhood and age of acquisition of the majority language, on heritage accent has been investigated in several bilingual dyads (e.g., Flores & Rato 2016; Kupisch et al. 2014, 2020; Loyd-Smith et al. 2020; Stangen et al. 2015). For example, Kupisch and colleagues report a positive correlation between heritage language experience during childhood and perceived nativeness in heritage Italian accent (Lloyd-Smith et al. 2020) and heritage Turkish accent (Kupisch et al. 2020). Studies that investigate the effect of more language experience through remigration to the homeland in adulthood show that its impact is not as strong as learning the heritage language in the host country during early years (Flores & Rato 2016). Kupish et al. (2020) and Stangen et al. (2015) examined age of onset of acquisition of the majority language and observe that this variable plays a role in heritage speech only if it is acquired past a certain age, since an adjustment tends to occur between the perceived degree of nativeness of the bilinguals' two languages, that is, sounding more native-like in one language leads to sounding less native-like in the other.

To the best of my knowledge, three of the first studies that investigated Portuguese heritage speech (i.e., Flores & Rato 2016; Flores et al. 2017; Rato et al. 2015) focused on exploring the apparent advantage of early acquisition and exposure in the speech performance of bilingual speakers in comparison with homeland speakers and advanced L2 learners.

Rato et al. (2015) investigated the heritage speech of German–European Portuguese bilinguals who were born and lived in Germany and Switzerland. Portuguese was the predominant language at home and all heritage speakers grew up with daily exposure to the heritage language spoken within the family. Three measures of perceived speech performance were collected, i.e., degree of nativeness in their accent, speech comprehensibility, and intelligibility. Semi-spontaneous speech productions of 24 EP speakers (12 heritage speakers, six homeland speakers, and six German L2 proficient Portuguese learners) were evaluated by 45 non-heritage homeland Portuguese raters. The group of raters were all born and raised in Portugal. The speech samples in Portuguese were collected in production tasks such as a semi-structured biographic and sociolinguistic oral interview and a story narration. Three sentences from each of the 24 speakers were selected based on the following criteria: Samples consisting of (i) full sentences with (ii) no morphological, syntactic, or lexical deviations, (iii) no hesitations, (iv) no long pauses (<0.3 sec), and (v) no cultural information. The three sentences were concatenated with a two-second inter-stimulus interval (ISI) to provide sufficient segmental and suprasegmental information for the ratings. The accent, comprehensibility, and intelligibility rating tasks included the same stimulus set. The global accent rating task used a two-step procedure. First, raters had to judge the speech samples binarily as 'native' or 'non-native' speech. Then, they had to indicate the degree of certainty of their assessment in a three-point scale (where 1 indicated 'certain', 2 'semi-certain', and 3 'uncertain'). This scale was then converted into a 6-point Likert scale by combining the two judgments (where 1 indicated 'certain of native speech', and 6 'certain of non-native speech'). The design of this task aimed at determining whether homeland speakers of European Portuguese perceived the accent of Portuguese heritage bilinguals who lived in a German-speaking country as (i) similar to the accent of Portuguese monolingual speakers or (ii) more similar to the accent of highly proficient German speakers of Portuguese L2. In the comprehensibility task, raters were asked to assess whether the sentences were easily understood in a 6-point scale ranging from 1 (very easy to understand) to 6 (very difficult to understand). The goal of this task was to examine whether the perceived degree of speech comprehensibility varied across the three participant groups and if there was an association

between these judgments and the degree of global accent. The intelligibility test was a gap-filling task. The sentences, from which three content words such as verbs, nouns, adjectives, and adverbs were omitted, were presented orthographically and participants were asked to fill in the gaps. The results of the accent rating task showed that the group of advanced L2 learners (L2ers) were perceived as non-native speakers, and heritage bilinguals were globally identified as native speakers of European Portuguese (M_{HS}: 1.66; M_{L2er} =5.82; M_{NS}=1.06, on a 9-point scale where 1 represents the most native-like). This finding suggests that early exposure to the HL can be a strong predictor of native accent development. However, individual variability in their speech ratings was also observed, with two HSs' judgments showing less consistency and lower degrees of certainty of native Portuguese accent (see Figure 18.1).

In addition, though the speech of all speaker groups was rated as highly comprehensible, a significant difference between the proficient L2 learners and the two other groups of EP speakers further revealed that the HS group was assessed similarly to the homeland group, and that comprehensibility and accent were positively correlated. Intelligibility ratings also yielded similar results for bilingual and monolingual speakers, that is, the speech of HSs was evaluated as being overall intelligible.

To investigate whether changes in the linguistic environment through remigration to the heritage country have an effect on heritage Portuguese speech, Flores and Rato (2016) examined the perceived global accent of 20 German-Portuguese bilingual 'returnees', who were raised in Germany and Switzerland but returned to Portugal, their parents' home country, after puberty. Following the same method (i.e., holistic accent ratings) as described in Rato et al. (2015), the researchers compared the HSs' accent to that of five highly proficient German speakers of L2 Portuguese and five homeland European Portuguese speakers. The group of heritage 'returnees' differed with respect to four variables, namely, age at immigration, age at remigration to heritage country, length of residence in their parents' homeland after remigration (i.e., LOR in Portugal), and length of residence in the German-speaking country (i.e., LOR in the host country). Although it was a heterogeneous group of heritage speakers, the ratings of 30 native European Portuguese listeners showed that the HSs accent fell within the homeland speakers' rating range but presented

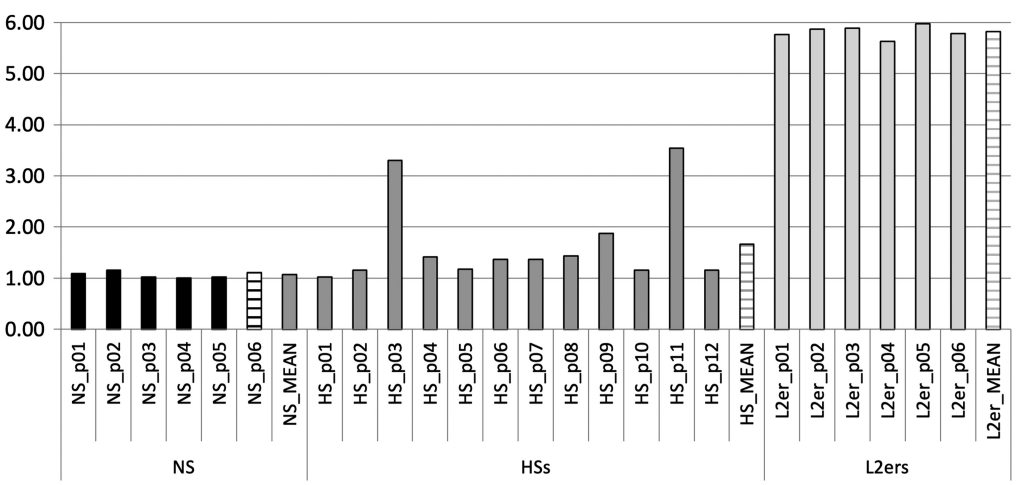

Figure 18.1 Individual accent ratings per group of speakers: HS=heritage speakers; L2er=L2 learners; NS=native homeland speakers (Rato et al. 2015).

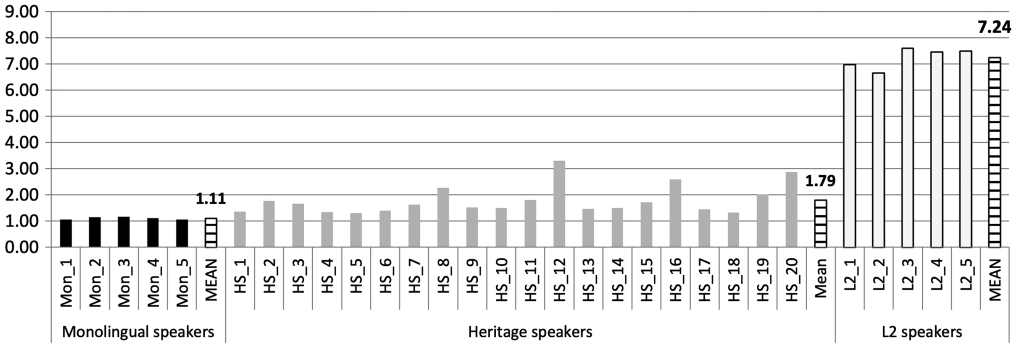

Figure 18.2 Individual accent ratings per group of speakers: HS=heritage speakers; L2=L2 speakers; Mon=native monolingual speakers (Flores & Rato 2016).

more individual variability, being perceived as more similar to monolinguals than to L2 speakers. The holistic accent ratings of semi-spontaneous speech samples revealed that HSs accent was perceived as closer (i.e., more native-like) to the monolingual speakers than to the L2 learners (M_{HS}: 1.79; M_{L2er} =7.24; M_{NS}=1.11, on a 9-point scale where 1 represents the most native-like). Compared to the findings reported by Rato et al. (2015), the accent ratings of HSs showed more variation, with some speakers being perceived as less native (see Figure 18.2).

These findings suggest that even when perceiving non-native (i.e., segmental and prosodic) patterns in the HSs' speech, Portuguese raters still significantly distinguish their heritage accent from the non-native accent of German L2 proficient speakers of Portuguese. In addition, further statistical analyses showed that from several possible factors impacting accent ratings (length of residence in Portugal before and after migration), the age at immigration to the German-speaking country (i.e., age of arrival – AOA) was the only significant variable that correlated with perceived degree of native-like accent. This finding suggests that early acquisition of and exposure to the heritage language sound system during childhood results in a perceived native-like speech. Spending their childhood in a German-dominant country did not appear to noticeably impact their heritage speech. The results also suggest that, given that most of the HSs (n=17) were returnees living in Portugal at the time of the study, recent remigration to Portugal and immersion experience with their heritage language may have resulted in a closer patterning to the homeland speakers. However, the detection of non-native patterns may persist in the returnees' speech over their lifespan. Therefore, further research with returnees with longer LOR in their heritage country is needed.

To further investigate whether the advantage of early acquisition is equally manifested in the phonological and morphosyntactic domains, Flores et al. (2017) compared the linguistic performance of German-Portuguese bilinguals with German learners of L2 Portuguese and homeland European speakers in the heritage language. The findings of the two experimental studies, one focused on the morphosyntactic knowledge of clitic object pronouns and the other on perceived global accent of Portuguese (whose experiment had the same design as the accent rating task described in Rato et al. 2015) confirmed the hypothesis that L2ers and HSs behave differently with respect to language properties that are acquired implicitly through early and extensive exposure to the minority language in early childhood. Although heritage bilinguals displayed more variability in their Portuguese speech than homeland speakers, the majority were rated as native speakers, being clearly set apart from the L2ers, who were consistently perceived as having a non-native

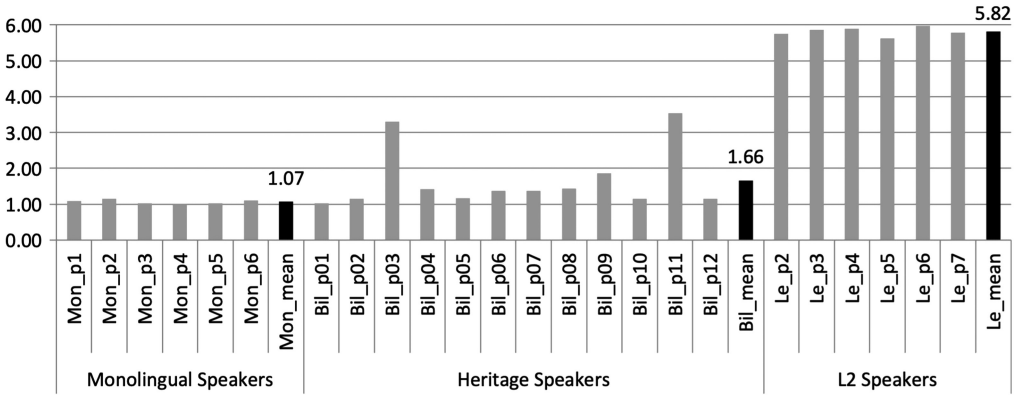

Figure 18.3 Individual accent ratings per group of speakers: Bil=bilingual speakers; Le=L2 speakers; Mon=native monolingual speakers (Flores et al. 2017).

accent. As reported in the two previous studies (Flores & Rato 2016; Rato et al. 2015), the divergence observed in the perceived accent of heritage speakers is generally observed in a few speakers who are rated as sounding less native (see Figure 18.3).

Despite superficial similarities reported in the morphosyntactic study between the L2ers and the HSs, the findings also showed that the heritage speakers' morphosyntactic competence is mainly shaped by their dominant exposure to colloquial Portuguese and reduced contact with formal registers. Though more research comparing the performance of bilingual speakers in different language domains is needed, these results suggest that morphosyntactic structures may show more deviation from the homeland baseline than their speech output (Polinskly 2018). These findings also indicate that intensive use of the home language within Portuguese migrant families predicts the development of an accent that is perceived as native-like. Unlike the studies that observe significant divergence in HSs' speech compared to native homeland speakers, the results of this study confirm that a largely native-like development of Portuguese HSs' speech can be explained by the speakers' degree of contact with Portuguese. All participants came from first-generation families, who spoke almost exclusively Portuguese at home, which was a strong indicator of native-like development, not only in the phonetic-phonological domain, but also in morphosyntax, despite the dominant contact with German.

A recent study with child heritage speakers (Correia et al. 2022) examined the perceived global accent of European Portuguese by 45 bilingual children (6 to 10 years of age) living in Germany and France, and 20 monolinguals living in Portugal. Their findings show that the accent spoken by EP-French and EP-German bilinguals is perceived as different from that spoken by monolinguals, suggesting that the exposure to a language from birth does not necessarily lead to a monolingual-like (i.e., non-heritage) accent. However, like research with adult heritage speakers, their results also show that there is much individual variability, with bilingual children being rated in a continuum that ranged from sounding native-like to sounding heavily foreign accented. This was significantly predicted by the children's productive vocabulary knowledge. The larger their productive lexical repertoire, the more likely were children perceived as having a native-like accent.

Research on heritage Portuguese accent has, so far, confirmed the outcomes observed in several other dominant-heritage languages dyads, in particular the intermediate patterning between

homeland native speakers and L2 learners and a wide range of individual variability (e.g., Chang & Yao 2016; Kupisch et al. 2014). As pointed out by Chang (2021), given the observed divergence in the accent of some heritage speakers from native monolinguals, future research could benefit from a more individual-centric approach in which variability is not reduced but rather accounted for statistically. See, for example, Chang and Yao (2024) for an individual-differences approach to analyzing variation among heritage speakers of Mandarin using a multiple factor analysis.

2.2 Heritage speech perception and production

Previous studies examining the phonological acquisition of HLs have found that exposure to a minority language in early childhood is reflected in an advantage in both production and perception of that language in adulthood in comparison to L2 learners, even in cases of prolonged loss of contact with the HL (e.g., Flores & Rauber 2011; Knightly et al. 2003; Oh et al. 2003). However, research findings on global perceived heritage accent also indicate that the phonetic and phonological patterns in heritage speech may diverge from that of the homeland due to language contact-induced changes. As suggested by Montrul (2016), the specific phonological and phonetic aspects that are affected in production by cross-linguistic influence need to be examined. In other words, to understand the degree to which the sound system of the dominant language influences heritage speech, the non-native features perceived in heritage Portuguese accent, as well as the perceptual categorization patterns in the processing of heritage speech, need to be identified. Similar to research in other heritage-dominant language pairings, speech production and perception has been investigated in the acquisition of heritage Portuguese, albeit very scarcely (e.g., Osborne 2021; Rato & Machado 2024).

Although heritage speech perception has not been investigated as much as heritage speech production, the existing body of work indicates that, in perception, heritage speakers display superior abilities in distinguishing sound contrasts that may be difficult for L2 learners (Polinsky 2018). It also suggests that early exposure to the heritage language may have a stronger effect on heritage language speech perception than in speech production, though it may not be sufficient for robustness in either speech modality (Kim, forthcoming). Although heritage speakers generally distinguish sound contrasts with ease, similarly to the findings on global qualities of heritage speech, research has shown that not all heritage speakers perform like native homeland speakers in the perception of phonemic contrasts. Some phonemic contrasts may be more difficult to perceive if there is a reduction of heritage language exposure at an early age (Ahn et al. 2017) or if the sound contrasts are in a position where the distinction is less acoustically salient, such as in word-final position (Lukyanchenko & Gor 2011).

Two studies have investigated heritage perception in Portuguese. Osborne's (2021) findings suggest that there may be differences in perception abilities that entail different levels of processing (phonetic and phonological), and that the degree of language dominance affects speech performance. The study investigated the perception of Brazilian Portuguese (BP) mid vowel contrasts (/ɛ/-/e/ and /ɔ/-/o/) by heritage speakers of Brazilian Portuguese who grew up in the United States. The perception data of 12 HSs of BP were collected in two experiments, an AXB discrimination test and a two-alternative forced-choice identification test (2AFC). In the discrimination test, participants heard three consecutive words and were asked to choose the word that sounded like the word in the middle (X). The identification task was divided into two blocks, with listeners hearing variations of the /ɔ/-/o/ continuum in the first block, and variations of the /ɛ/-/e/ continuum in the second block. In each block, participants were asked to match the word they heard with one of two pictures. All heritage speakers discriminated the mid vowel Portuguese contrasts (/ɛ/-/e/

and /ɔ/-/o/), but, at the phonological level, they did not perform as consistently. Specifically, the high-English-dominant bilingual group did not identify the back vowels as categorically as the low-English-dominant speakers, suggesting that aspects of HL phonology may be affected by the HSs degree of language dominance. The author mentions that this type of asymmetry in perception between mid-front and mid-back vowels has been found in other studies examining these contrasts (e.g., Amengual 2015; Feiden et al. 2014).

To investigate cross-linguistic influence on heritage speech and the extent to which bilingual speakers maintain language-specific phonetic contrasts, Rato and Machado (2024) analyzed the production and perception of a shared set of phonemes that differ acoustically in voice onset time across the two languages of the bilingual dyad (heritage-Portuguese and dominant-English). Thus, it investigated whether English-Portuguese bilinguals are able to distinguish and produce the language-specific phonetic realizations of voiceless stops. Specifically, the study examined whether bilingual speakers are able to suppress an allophonic rule of the dominant language (i.e., the realization of aspirated voiceless stops in syllable onset position) when perceiving and producing their heritage language; and if the short-lag voicing in the speakers' heritage language affects the discrimination and production of the same phonemes in their majority language. To address these research questions, a combination of experimental tasks with a cross-linguistic design was applied. A total of 12 early bilingual speakers of second-generation European Portuguese (EP) heritage who were born in Toronto, Canada, to first-generation Portuguese parents and were exposed to Portuguese from birth participated in this study. In addition, a group of four native speakers of Canadian English (CE) and four native speakers of EP, residing in the homeland, provided monolingual baseline data and a group of 22 native EP speakers participated as raters. The data were collected through a set of production and perception tasks, namely, a pseudo-word reading task, a rated dissimilarity task (RDT), and a perceptual assimilation (PAT) task. In the task of rated dissimilarity (RDT), the listeners evaluated the degree of (dis)similarity between two instances of stop realizations presented in pseudo words, on a 7-point scale (from 1=same to 7=different). The pairs included same-consonant tokens in CE-CE pairs and in EP-EP pairs, which were used as controls, and the target trials included same consonant CE-EP sequences. In the cross-language perceptual assimilation task (PAT), participants listened to pseudo words, with both Portuguese and English stop consonants /p/, /t/, /k/, /b/, /d/, /g/ and categorized them as one of the stops, using orthographic response labels <p, t, k, b, d, g>, or as a 'no fit'. After each token, participants were also asked to evaluate their representativeness as exemplars of the selected category, i.e., to rate category goodness-of-fit (GoF) for each stop realization according to the phonological categories of their dominant language, in a 7-point Likert scale ranging from 1 (poor exemplar) to 7 (very good exemplar). The pseudo-word reading tasks in Portuguese and English included 24 pseudo disyllabic words each, in which the target consonants were presented in stressed syllable-initial position. Participants were asked to read the pseudo-Portuguese words (with a V.CV syllable structure) embedded in a carrier sentence (e.g., "Diz [a.Ci] para ti") and 24 pseudo-English words also in a carrier sentence (e.g., 'I shout [a.Ci] to you'). Results revealed that bilingual CE-EP speakers produced language-specific realizations of aspirated English and unaspirated Portuguese voiceless stops in syllable onset position (see Figure 18.4).

Bilingual speakers were able to reset the phonetic boundaries of the voiceless stops cross-linguistically by producing the target consonants with significantly different VOTs in the two languages of the bilingual dyad. Furthermore, stop production in their heritage and dominant languages did not diverge from homeland baseline data, that is, English-Portuguese bilinguals matched the VOTs of monolingual speakers. This finding showed that the group of bilinguals displayed native-like patterns in both languages, as observed in previous research (e.g., Antoniou

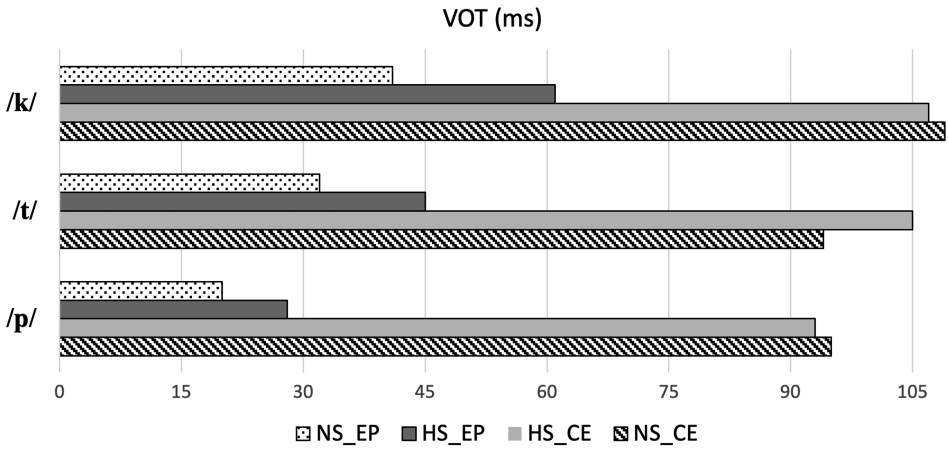

Figure 18.4 Average VOT (ms) across groups: NS_EP=monolingual Portuguese stops; NS_CE=monolingual English stops; HS-EP=bilingual Portuguese stops; HS_CE=bilingual English stops (Rato & Machado 2024).

et al. 2010; Kang & Guion 2006; Kim 2012), which suggests that they have developed separate language-specific phones for the aspirated-unaspirated cross-linguistic contrasts and were able to suppress cross-linguistic influence in the production of both languages. However, these cross-linguistic acoustic differences were disregarded in perception with the bilingual speakers not distinguishing between aspirated English and plain Portuguese voiceless stops, which were previously validated by two groups of monolingual listeners of each language. These results indicate that the bilingual group did not rely on the low-level acoustic differences, but rather on their similar phonemic features, thus assimilating them as the same phonological categories. The findings therefore showed an asymmetry with respect to HSs speech perception and production performance, in which cross-linguistic (non-contrastive) acoustic differences were disregarded in perceptual categorization but emphasized in production. The researchers explain that one possible reason for this asymmetry could be related to the methodological procedure adopted in this study. Particularly, the simultaneous oral presentation of English and Portuguese stimuli in the perception tasks that may have triggered both language modes. Instead of activating the low-level phonetic processing of each language separately, it triggered a 'good-enough' use of phonological knowledge in which cross-linguistic assimilation was observed, leading bilinguals to ignore non-distinctive contrasts (Polinsky 2018). In production, however, cross-linguistic dissimilation was observed, with bilingual speakers producing language-specific voiceless stop phones, which seems to have been enhanced with a shift in language mode (from Portuguese to English) during data collection. Future research on heritage Portuguese perception and production is critically needed to identify the features of heritage accent (vowels, consonants, intonation, speech rate) and to examine the relationship between these given features and heritage accent. Rato and Machado (2024) found a positive correlation between degree of VOT for alveolar /t/ and perceived accent rating with the HSs who produced the alveolar stop with a shorter VOT, rated as having a more native-like accent. However, as in other studies (e.g., Au et al. 2002; Oh et al. 2003), this relationship was not consistent across all sounds. For instance, Lein and colleagues (2015) did not find any systematic relationship between VOT and perceived heritage accent, with some speakers producing stop consonants with VOTs within monolingual

range and being perceived to sound non-native-like, while others displayed deviant VOTs and sounded native-like. These findings indicate that when native raters evaluate heritage speakers' accent, their attention is not restricted to a given phonetic feature; rather, there may be other aspects contributing to heritage accent, including at the suprasegmental level. Further research is needed to disentangle the relative role of different aspects of speech contributing to the perception of a heritage accent.

3 Concluding remarks

There is growing evidence that heritage speakers' sound systems differ from those of monolingual native speakers. However, as suggested by Kim (forthcoming), the extent to which HL speech and the baseline overlaps and/or differentiates is still not clear and further studies on HL phonology are necessary. This is particularly critical for heritage Portuguese phonology, for which research is scarce. As the present overview indicated, there are very few studies in this area and, overall, the study of the phonological acquisition of heritage Portuguese is an open area of investigation.

Nevertheless, the existing research on the phonological acquisition of Portuguese as a heritage language provides some relevant findings. First, research suggests that early and regular exposure to the heritage language has lasting effects on the development of an accent that is perceived by homeland speakers as native. The findings to this date also point to an intermediate patterning in HSs accent between baseline monolingual speakers and advanced L2 learners, explained by the behavior of some HSs that sound less native than others. Therefore, an individual-differences approach to the analysis of heritage speech is deemed more suitable to capture the variability among heritage speakers with similar linguistic background and language acquisition history (Chang & Yao 2024). Research in this field suggests that several factors play a role in heritage speech outcomes, including linguistic factors (e.g., orthography, dialectal variation, etc.) and extralinguistic variables (e.g., age at exposure to the majority language, heritage language use, ethnic orientation, heritage language proficiency). However, in the current body of empirical research on Portuguese heritage speech, the relative weight of each factor is still an open question to be explored. Flores and Rato (2016), for instance, report that age of arrival (AOA) to the host country was the only significant factor among a set of variables that was correlated with degree of perceived global accent, and Correia et al. (2022) report that only productive lexical knowledge predicted individual variability in child heritage accent. But see Umbal (2023), for example, who adopts a comparative variationist approach to examine the interplay between language internal-, language contact-, and ethnic identity-accounts underlying the phonetic variation in the speech of two generations of Tagalog heritage speakers in Canada. Therefore, the interplay of different factors needs to be further explored, as well as the role of cross-linguistic influence on both heritage and dominant languages. Overall, work on the HSs majority language is currently lacking (Chang 2021) when compared to the number of studies on the minority language. In the case of Portuguese, only Rato and Vanina (2024) provide acoustic measures of speech production (e.g., VOT of voiceless stops in both the dominant English and the heritage Portuguese language), but no global measures of accent in both languages in the majority-minority dyad have been analyzed. Therefore, additional work on the HSs dominant language is pressing to further describe and understand the unique profile of bilingualism (Chang 2021). Future research on both speech domains is also critically needed to understand how perception and production are related in heritage speech, specifically if one modality leads or lags the other in heritage speech development.

There is also a need for studies that focus on the perception and production of suprasegmental phonology, for which there is no research on heritage Portuguese. Although prosody plays an

important role in the perception of heritage accent (e.g., Shin 2005), it is still unclear whether it plays a bigger role than segments. Some topics for investigation include heritage Portuguese intonation and stress so as to test whether these suprasegmental features are more subject to change and variation, as observed by Polinsky (2018) and in the study of speech acquisition and development of two closely related varieties of Portuguese (e.g., Brazilian and European Portuguese), which differ in speech rhythm (Frota & Vigário 2001). Early heritage language experience has been highlighted as the main predictor of heritage language outcomes. It is therefore relevant to conduct longitudinal research to examine HSs over time in comparison to their homeland counterparts or cross-sectional studies to compare HSs of different age groups to further understand the development of heritage speakers' phonetic and phonological systems during their lifespan. It is also pressing to examine the developmental trajectory of heritage speech in child heritage populations to further understand when the heritage accent emerges, how heritage speech develops over time, and the effect of language dominance shift on its development.

All these empirical gaps in the acquisition of Portuguese heritage speech are clear directions for future research. The investigation of the heritage Portuguese phonological system has important implications and is needed not only to further the understanding of HL acquisition, but also for educational practices.

Note

1 Countries where Portuguese is an official language: Portugal, Brazil, Angola, Cabo Verde, Guinea-Bissau, Equatorial Guinea, Mozambique, São Tomé and Príncipe, East Timor, and Macau (China).

Further reading

Chang, Charles. 2021. Phonetics and phonology of heritage languages. In Silvina Montrul and Maria Polinsky (eds.), *The Cambridge Handbook of Heritage Languages and Linguistics*. Cambridge: Cambridge University Press, 581–612.

Kim, Ji Young. Forthcoming. The phonetics and phonology of heritage language speakers. In Mark Amengual (ed.), *The Cambridge Handbook of Bilingual Phonetics and Phonology*. Cambridge: Cambridge University Press.

Polinsky, Maria. 2018. *Heritage Languages and Their Speakers*. Cambridge: Cambridge University Press.

Rato, Anabela and Cristina Flores. Forthcoming. The acquisition of Portuguese as a heritage language. In Ana Carvalho and Livia Oushiro (eds.), *The Oxford Handbook of the Portuguese Language.* Oxford: Oxford University Press.

References

Ahn, Sunyoung, Charles B. Chang, Robert DeKeyser, and Sunyoung Lee-Ellis. 2017. Age effects in first language attrition: Speech perception by Korean-English bilinguals. *Language Learning* 67.694–733.

Altenhofen, Cléo V. 1996. *Hunsrückisch in Rio Grande do Sul. Ein Beitrag zur Beschreibung einer deutsch-brasilianischen Dialektvarietät im Kontakt mit dem Portugiesischen*. Stuttgart: Steiner.

Amengual, Mark. 2015. The perception of language-specific phonetic categories does not guarantee accurate phonological representations in the lexicon of early bilinguals. *Applied Psycholinguistics* 37.1–31.

Antoniou, Mark, Catherine Best, Michael D. Tyler, and Christian Kroos. 2010. Language context elicits native-like stop voicing in early bilinguals' productions in both L1 and L2. *Journal of Phonetics* 38.640–653.

Au, Terry Kit-Fong, Leah M. Knightly, Sun-Ah Jun, and Janet S. Oh. 2002. Overhearing a language during childhood. *Psychological Science* 13.238–243.

Au, Terry Kit-Fong and Janet S. Oh. 2009. Korean as a heritage language. In Ping Li (ed.), *Handbook of East Asian Psycholinguistics, Part III: Korean Psycholinguistics*. Cambridge: Cambridge University Press, 268–275.

Au, Terry Kit-Fong, Janet S. Oh, Leah M. Knightly, Sun-Ah Jun, and Laura F. Romo. 2008. Salvaging a childhood language. *Journal of Memory and Language* 58.998–1011.

Baker, Wendy and Pavel Trofimovich. 2005. Interaction of native- and second-language vowel system(s) in early and late bilinguals. *Language and Speech* 48.1–27.

Benmamoun, Elabbas, Silvina Montrul, and Maria Polinsky. 2010. Prolegomena to heritage linguistics [white paper]. University of Illinois at Urbana-Champaign and Harvard University. [Online] Available at: https://www.semanticscholar.org/paper/Prolegomena-to-Heritage-Linguistics-Benmamoun-Montrul/4fe2a88fd8a54f1f0923e9a1b512a71366a8fb3e.

Benmamoun, Elabbas, Silvina Montrul, and Maria Polinsky. 2013. Heritage languages and their speakers: Opportunities and challenges for linguistics. *Theoretical Linguistics* 39.129–181.

Castro, Tammer, Jason Rothman, and Marit Westergaard. 2017. On the directionality of cross-linguistic effects on bidialectal bilingualism. *Frontiers in Psychology* 8.

Chang, Charles B. 2016. Bilingual perceptual benefits of experience with a heritage language. *Bilingualism. Language and Cognition* 19.791–809.

Chang, Charles B. 2021. Phonetics and phonology of heritage languages. In Silvina Montrul and Maria Polinsky (eds.), *The Cambridge Handbook of Heritage Languages and Linguistics*. Cambridge: Cambridge University Press, 581–612.

Chang, Charles B. and Yao Yao. 2016. Toward an understanding of heritage prosody: Acoustic and perceptual properties of tone produced by heritage, native, and second language speakers of Mandarin. *Heritage Language Journal* 13.134–160.

Chang, Charles B. and Yao Yao. 2024. An individual-differences perspective on variation in heritage Mandarin speakers. In Rajiv Rao (ed.), *The Phonetics and Phonology of Heritage Languages*. Cambridge: Cambridge University Press, 208–236.

Chang, Charles B., Yao Yao, Erin F. Haynes, and Russell Rhodes. 2011. Production of phonetic and phonological contrast by heritage speakers of Mandarin. *Journal of the Acoustical Society of America* 129.3964–3980.

Colantoni, Laura, Alejandro Cuza, and Natalia Mazzaro. 2016. Task related effects in the prosody of Spanish heritage speakers and long-term immigrants. In Meghan E. Armstrong, Nicholas Henriksen, and Maria del Mar Vanrell (eds.), *Intonational Grammar in Ibero-Romance: Approaches Across Linguistic Subfields*. Amsterdam: John Benjamins, 1–24.

Correia, Liliana, Chao Zhou, and Cristina Flores. 2022. Productive lexical knowledge predicts bilingual children's global accent in their heritage language. Paper presented at the 15th Generative Approaches to Bilingualism (GALA 15). Goethe-Universität Frankfurt, Germany.

Costa, João, Maria Lobo, and Fernanda Pratas. 2016. Clitic production by Portuguese and Capeverdean children: Omission in bilingualism. *Probus* 28.271–291.

De Leeuw, Esther, Monika S. Schmid, and Ineke Mennen. 2010. Perception of foreign accent in native speech. *Bilingualism: Language and Cognition* 13.33–40.

Engel de Abreu, Pascale M. J., Anabela Cruz-Santos, Carlos J. Tourinho, Romain Martin, and Ellen Bialystok. 2012. Bilingualism enriches the poor: Enhanced cognitive control in low-income minority children. *Psychological Science* 23.1364–71.

Escudero, Paola, Paul Boersma, Andreia Rauber, and Ricardo Bion. 2009. A cross-dialect acoustic description of vowels: Brazilian and European Portuguese. *The Journal of the Acoustical Society of America* 126.1379–1393.

Feiden, Juliana A., Ubiratã K. Alves, and Ingrid Finger. 2014. O efeito da anterioridade e da altura na identificação das vogais médias altas e médias baixas do português brasileiro por falantes de espanhol. *Letras de Hoje* 49.85–94.

Ferreira, Helena. 2020. *Como desenvolver a consciência linguística relativamente à existência de interferências do Inglês no Português Língua Segunda / Língua de Herança?* Master's thesis, University of Porto.

Figueira, Fernanda. 2017. *Conhecimento lexical de crianças bilingues falantes de português europeu e português brasileiro*. Master's thesis, Instituto Politécnico de Setúbal.

Flores, Cristina. 2010. The effect of age on language attrition: Evidence from bilingual returnees. *Bilingualism. Language and Cognition* 13.533–546.

Flores, Cristina, Pilar Barbosa, and Manuela Casa Nova. 2016. A closer look at crosslinguistic influence in the acquisition of Portuguese as a heritage language. In Sambor Grucza, Magdalena Olpinska-Szkieko,

and Piotr Romanowski (eds.), *Bilingual Landscape of the Contemporary World*. Frankfurt/Main: Peter Lang Verlag, 75–94.

Flores, Cristina and Anabela Rato. 2016. Global accent in the Portuguese speech of heritage returnees. *Heritage Language Journal – Special Issue on Heritage Phonetics and Phonology* 13.161–183.

Flores, Cristina and Andreia Rauber. 2011. Perception of German vowels by bilingual Portuguese-German returnees: A case of phonological attrition? In Esther Rinke and Tanja Kupisch (eds.), *The Development of Grammar*: *Language Acquisition and Diachronic Change – Volume in Honor of Jürgen M. Meisel*. Amsterdam: John Benjamins, 287–305.

Flores, Cristina, Esther Rinke, and Anabela Rato. 2017. Comparing the outcomes of early and late acquisition of European Portuguese. *Heritage Language Journal* 14.124–149.

Flores, Cristina and Neal Snape. 2021. Language attrition and heritage language reversal in returnees. In Silvina Montrul and Maria Polinsky (eds.), *The Cambridge Handbook of Heritage Languages and Linguistics*. Cambridge: Cambridge University Press, 351–372.

Flores, Cristina, Chao Zhou, and Carina Eira. 2022. 'I no longer count in German'. On dominance shift in returnee heritage speakers. *Applied Psycholinguistics* 43.1019–1043.

Flores, Cristina, Esther Rinke, and Claudia Wagner. 2022. 'To hón ich imma insistieat.' Syntactic stability in heritage Hunsrückisch German spoken in Brazil. *Linguistic Approaches to Bilingualism* 12.251–279.

Flores, Cristina, Maria Lurdes Gonçalves, Esther Rinke, and Jacopo Torregrossa. 2022. Perspetivas múltiplas sobre a competência bilingue de crianças lusodescendentes residente na Suíça. A investigação linguística em diálogo com a didática. *Revista Portuguesa de Educação* 35.102–131.

Frota, Sónia and Marina Vigário. 2001. On the correlates of rhythmic distinctions: The European/Brazilian case. *Probus* 13.247–275.

Godson, Linda. 2004. Vowel production in the speech of Western Armenian heritage speakers. *Heritage Language Journal* 2.44–69.

Gor, Kira. 2014. Raspberry, not a car: Context predictability and a phonological advantage in early and late learners' processing speech in noise. *Frontiers in Linguistics* 5.

Guion, Susan G. 2003. The vowel systems of Quichua-Spanish bilinguals. *Phonetica* 60.98–128.

Halter, Marilyn. 2005. Cape Verdeans in the United States. In Melvin Ember, Carol Ember, and Ian Skoggard (eds.), *Encyclopedia of Diasporas: Diaspora Communities*. New York: Springer, 615–623.

Hopp, Holger and Monika S. Schmid. 2013. Perceived foreign accent in L1 attrition and L2 acquisition: The impact of age of acquisition and bilingualism. *Applied Psycholinguistics* 34.361–394.

Hrycyna, Melania, Natalia Lapinskaya, Alexei Kochetov, and Naomi Nagy. 2011. VOT drift in 3 generations of heritage language speakers in Toronto. *Canadian Acoustics* 39.166–167.

Jochmann, Layla C. 2020. *Português como língua de herança. A coesão em textos escritos por crianças plurilingues*. Berlin: Peter Lang.

Kan, Rachel. 2020. Suprasegmental and prosodic features contributing to perceived accent in heritage Cantonese. *Proceedings of the 10th International Conference on Speech Prosody*, 101–105.

Kang, Kyoung and Susan G. Guion. 2006. Phonological systems in bilinguals: Age of learning effects on stop consonant systems of Korean-English bilinguals. *Journal of the Acoustical Society of America* 119.1672–1683.

Kim, Ji Young. 2012. *Discrepancy between the Perception and Production of Stop Consonants by Spanish Heritage Speakers in the United States*. Master's thesis, Korea University.

Kim, Ji Young. 2015. Perception and production of Spanish lexical stress by Spanish heritage speakers and English L2 learners of Spanish. In Erik W. Willis, Pedro Martín Butragueño, and Esther Herrera Zendejas (eds.), *Selected Proceedings of the 6th Conference on Laboratory Approaches to Romance Phonology*. Somerville, MA: Cascadilla Proceedings Project, 106–128.

Kim, Ji Young. 2020. Discrepancy between heritage speakers' use of suprasegmental cues in the perception and production of Spanish lexical stress. *Bilingualism: Language and Cognition* 23.233–250.

Knightly, Leah M., Sun-Ah Jun, Janet S. Oh, and Terry Kit-Fong Au. 2003. Production benefits of childhood overhearing. *Journal of Acoustical Society of America* 114.465–374.

Koch, Walter. 1974. *Falares alemães no Rio Grande do Sul*. Porto Alegre: UFRGS.

Koven, Michèle E. J. 2007. *Selves in Two Languages: Bilinguals' Verbal Enactments of Identity in French and Portuguese*. Amsterdam: John Benjamins.

Kupisch, Tanja, Dagmar Barton, Ewgenia Klaschik, Tatjana Lein, Ilse Stangen, and Joost van de Weijer, 2014. Foreign accent in adult simultaneous bilinguals. *Heritage Language Journal*, 11.123–150.

Kupisch, Tanja, Tammer Castro, Martin Krämer, and Marit Westergaard. 2023. Phonological influence in bilectal speakers of Brazilian Portuguese and European Portuguese. *International Journal of Bilingualism* 0.1–18.

Kupisch, Tanja, Nadine Kolb, Yulia Rodina, and Olga Urek. 2021. Foreign accent in pre- and primary school heritage bilinguals. *Languages* 6.96.

Kupisch, Tanja, Anika Lloyd-Smith, and Ilse Stangen. 2020. Perceived global accent in Turkish heritage speakers in Germany. In Fatih Bayran (ed.), *Studies in Turkish as a heritage language*. Amsterdam: John Benjamins, 207–228.

Lebert, Joanne-M. 1999. *Negotiating Angolan-ness in Diaspora*. Master's thesis, University of Toronto.

Lein, Tatjana, Tanja Kupisch, and Joost van de Weijer. 2015. Voice onset time and global foreign accent in German-French simultaneous bilinguals during adulthood. *International Journal of Bilingualism* 20.732–749.

Lloyd-Smith, Anika, Marieke Einfeldt, and Tanja Kupisch. 2020. Italian-German bilinguals: The effects of heritage language use on accent in early-acquired languages. *International Journal of Bilingualism* 24.289–304.

Lukyanchenko, Anna, and Kira Gor. 2011. Perceptual correlates of phonological representations in heritage speakers and L2 learners. In Nick Danis, Kate Mesh, and Hyunsuk Sung (eds.), *Proceedings of the 35th Annual Boston University Conference on Language Development*. Somerville, MA: Cascadilla Press, 414–426.

Martins, Ana Maria. 2006. Emphatic affirmation and polarity: Contrasting European Portuguese with Brazilian Portuguese, Spanish, Catalan, and Galician. In Jenny Doetjes and Paz González (eds.), *Romance Languages and Linguistic Theory 2004*. Amsterdam: John Benjamins, 197–223.

Montrul, Silvina. 2010. Current issues in heritage language acquisition. *Annual Review of Applied Linguistics* 30.3–23.

Montrul, Silvina. 2016. *The Acquisition of Heritage Languages*. Cambridge: Cambridge University Press.

Nagy, Naomi. 2015. A sociolinguistic view of null subjects and VOT in Toronto heritage languages. *Lingua* 164.309–327.

Nagy, Naomi and Alexei Kochetov. 2013. Voice onset time across the generations: A cross-linguistic study of contact-induced change. In Peter Siemund, Ingrid Gogolin, Monika E. Schulz, and Julia Davydova (eds.), *Multilingualism and Language Contact in Urban Areas: Acquisition, Identities, Space, Education*. Amsterdam, Netherlands: John Benjamins, 19–38.

Nunes, Naidea N. 2018. A voz de origem: Identidade sociocultural e linguística nas histórias de vida de migrantes madeirenses e seus descendentes na Venezuela. *Pensardiverso: Revista de Estudos Lusófonos* 6.112–150.

Oh, Janet S., Sun-Ah Jun, Leah M. Knightly, and Terry Kit-Fong Au. 2003. Holding on to childhood language memory. *Cognition* 86.B53–B64.

Osborne, Denise Maria. 2021. Perception of Portuguese mid vowels by heritage speakers of Brazilian Portuguese. *Heritage Language Journal* 18.1–32.

Pereira, Bruna. 2017. *Estudo linguístico e sociocultural da emigração madeirense para a África do Sul a partir da década de 1960 (no âmbito do Projeto Nona Ilha)*. Master's thesis, University of Madeira.

Piippo, Jarna. 2016. *Línguas maternas no ensino básico: Espanhol e português na área metropolitana de Helsínquia*. Ph.D. dissertation, University of Helsinki.

Pires, Acrisio and Jason Rothman. 2009. Disentangling sources of incomplete acquisition: An explanation for competence divergence across heritage grammars. *International Journal of Bilingualism* 13.211–238.

Polinsky, Maria. 2018. *Heritage Languages and Their Speakers*. Cambridge: Cambridge University Press.

Polinsky, Maria and Olga Kagan. 2007. Heritage Languages: In the 'Wild' and in the classroom. *Language and Linguistics Compass* 1.368–395.

Rao, Rajiv. 2015. Manifestations of /bdg/ in heritage speakers of Spanish. *Heritage Language Journal* 12.48–74.

Rao, Rajiv and Rebecca E. Ronquest. 2015. The Heritage Spanish phonetic/phonological system: Looking back and moving forward. *Studies in Hispanic and Lusophone Linguistics* 8.403–414.

Rato, Anabela, Cristina Flores, Daniela Neves, and Diana Oliveira. 2015. A competência fonológica de falantes bilíngues luso-alemães. *Diacrítica* 29(1).297–326.

Rato, Anabela and Vanina Machado. 2024. Perception and production of English and Portuguese voiceless stops by heritage learners. In Rajiv Rao (ed.), *The Phonetics and Phonology of Heritage Languages*. Cambridge: Cambridge University Press. 343–360.

Ribeiro, Patrícia. 2021. Uso do artigo definido na construção do sintagma nominal: Interações entre o finlandês e o português como língua de herança. *O POLH na Europa. Sagarana Editora* 3.189–200.

Rinke, Esther and Cristina Flores. 2018. Another look at the interpretation of overt and null pronominal subjects in bilingual language acquisition: Heritage Portuguese in contact with German and Spanish. *Glossa: A Journal of General Linguistics* 3.68.

Rinke, Esther and Cristina Flores. 2021. Portuguese as Heritage Language in Germany—A Linguistic Perspective. *Languages* 6.10.

Ronquest, Rebecca E. 2013. An acoustic examination of unstressed vowel reduction in heritage Spanish. In C. Howe, S. E. Blackwell, and M. Quesada (eds.), *Selected Proceedings of the 15th Hispanic Linguistics Symposium*. Somerville, MA: Cascadilla Press, 157–171.

Rothman, Jason. 2009. Understanding the nature of early bilingualism: Romance languages as heritage languages. *International Journal of Bilingualism* 13.155–163.

Scetti, Fabio. 2016. Dialectal Variation of the Portuguese Language. Evolution of the Portuguese Language in the Context of Migration: The' comunidade portuguesa' of Montreal. *Limite. Revista de Estudios Portugueses y de la lusofonia* 10.107–119.

Shin, Eurie. 2005. The perception of foreign accents in spoken Korean by prosody: Comparison of heritage and non-heritage speakers. *The Korean Language in America* 10. 103–118.

Silva, Gláucia and Cristiane Soares. 2021. "Uma problema complicado": o gênero gramatical em produções semiespontâneas de aprendizes de português. In Mauricio J. Souza Neto and Nilma N. Dominique (eds.), *Microgeopolítica da Língua Portuguesa: Ações, desafios e perspectivas*. Roosevelt: Bovista Press, 109–130.

Simard, Daphnée, Véronique Fortier, and Denis Foucambert. 2013. Measuring metasyntactic ability among heritage language children. *Bilingualism: Language and Cognition* 16.19–31.

Stangen, Ilse, Tanja Kupisch, Ergün Proietti, and Marina Zielke. 2015. Foreign accent in heritage speakers of Turkish in Germany. In Hagen Peukert (ed), *Transfer Effects in Multilingual Language Development*. Amsterdam: Benjamins, 87–108.

Tomaz, Margarida, Maria Lobo, Ana Madeira, Carla Soares-Jesel, and Stephanie Vaz. 2019. Omissão e colocação de clíticos por crianças bilíngues Português-Francês. *Revista da Associação Portuguesa de Linguística* 5.385–412.

Tomaz, Margarida, Maria Lobo, Ana Maria Madeira, Carla Soares-Jesel, and Stephanie Vaz. 2020. Efeitos de dominância linguística e de tempo de exposição formal à língua na produção de pronomes clíticos por crianças bilíngues português/francês. *Revista da Associação Portuguesa de Linguística* 7.350–367.

Umbal, Pocholo. 2023. *A Comparative Variationist Analysis of Phonetic Variation and Change in Toronto Heritage Tagalog*. Ph.D. dissertation, University of Toronto.

Vago, Robert M. 1991. Paradigmatic regularity in first language attrition. In Herbert W. Seliger and Robert M. Vago (eds.), *First Language Attrition*. Cambridge: Cambridge University Press, 241–252.

Wrembel, Magdalena, Marta Marecka, Jakub Szewcyk, and Agniezka Otwinowska. 2019. The predictors of foreign-accentedness in the home language of Polish–English bilingual children. *Bilingualism: Language and Cognition* 22.383–400.

Yang, Bei. 2015. *Perception and Production of Mandarin Tones by Native Speakers and L2 Learners*. Berlin: Springer.